

Advanced signal processing

Exam 2

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1 Introduction

This lab journal will take a shallow dive in to adaptive signal processing, specifically comparison of three common types of implementation, the regular *LMS*(REF), the *leaky-LMS*(REF) and the *Variable step-size LMS*(REF). Comparison of their performances in an experimental sense, a delve into their specifics, use cases and a critical discussion.
It is a assumed that the reader has some basic understand of digital filtering and a general grasp of their implementation, therefore a minimum amount of effort will be spend on that topic, however characterics regarding these mentioned types of filters, will be explained and discussed.
Filtering in general, is the art of removing the unwanted or highlighting and improving specific feature of input signals. Mainly in the analog domain, a filter would be designed for given frequency or bandwidth, with little possibility of expanding or changing. Such is the same problem when designing FIR(REF) or IIR(REF) filters in the digital domains. The lack of ability to adapt the characterics of the filtering gave birth to the aptly named *Adaptive filter*.

2 Theoretical background

Contrary to regular filtering, the designer targets a specific frequency or bandwidth, placing zeroes and poles on the unit-circle to *approximate*, a desired result, or atleast get somewhat close to something acceptable.

In adaptive filtering, we're more concerned about the system as an input and how we implement an algorithm to force the error to be zero. In this lab journal the overall system we'll be examining will take the form of:

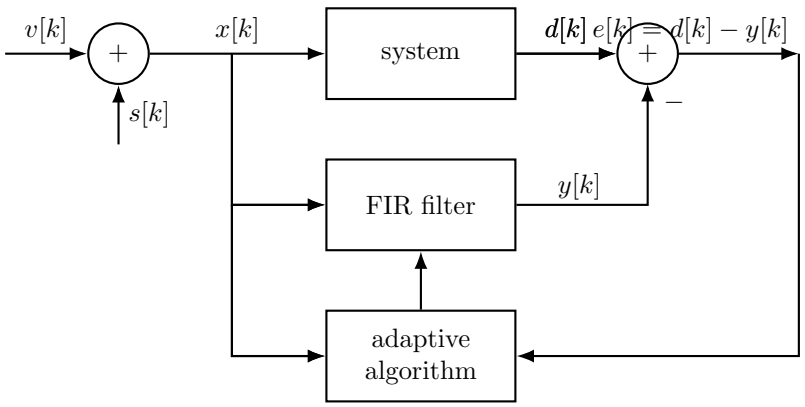


Figure 1: Adaptive FIR filter structure.

Where $x[k]$ will be some randomly generated noise, with a zero-mean, unit variance, passing through a system summed together with a sinusoid $s[k]$, for testing purposes, to experimentally see how the system responds to both noise and a signal combined.

The simple LMS filter works by taking the output of the system, as the desired output $d[k]$ and then subtracting $y[k]$, which is the output of the FIR filter, governed by a vector of weights $w[k]$. The The LMS algorithm will then try to force the error $e[k] = d[k] - y[k] \approx 0$ by tuning the weights in descrete steps.
The vector $\mathbf{W}[k]$ hold the coefficients and the vector $\mathbf{X}[k]$ holds old samples :

$$\mathbf{W}[k] = \begin{bmatrix} w_0[k] \\ w_1[k] \\ ... \\ w_N - 1[k] \end{bmatrix}, \qquad \mathbf{X}[k] = \begin{bmatrix} x[k] \\ x[k-1] \\ ... \\ x[k-N+1] \end{bmatrix}$$

The complete algorithm for computing the coefficiens for the vector:

$$\mathbf{W}(k+1) = \mathbf{W}[k] + 2\mu e[k]\mathbf{x}[k]$$

The factor μ is factor that gives some control of how aggressize the filter should act, as it a scalar that is multiplied onto the weights and the old samples. The idea of the algorithm is then, that each iteration that it runs, the weights

of the filter are multipled with the signal, which affects how $y[k+1]$ samples will behave, this affects the error, and it repeats. Each iteration the weights gets updates aswell, so that they are applied with a different value at the next iteration.

3 Numerical experiment

3.1 LMS experimentation

Looking at the equations for the LMS filter, we're given some handles to tune the filter with namely μ, M the amount of filter taps and N how long the algorithm will run.

For the initial tests, we'll keep the amount of filter tabs fixed to $M = 3$, and we'll instead vary the step size by a factor of 1000, and keep the test running for a 1000 samples.

3.2 leaky LMS experimentation

3.3 variable step-size experimentation

4 Results

4.1 Analysis

4.2 Discussion

5 Conclusion